

A Lamport-Style Proof Package for the Bound $g(n) \geq \lceil 7n/27 \rceil$

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Abstract

A Euclidean minimum-distance graph is the unit-distance graph of a finite point set in $\mathbb{R}^2$ whose pairwise distances are all at least 1. Let $g(n)$ be the minimum, over all such graphs on n vertices, of the independence number. This document gives a complete Lamport-style proof, relative to Swanepoel's published broken-lattice inputs, that every Euclidean minimum-distance graph G satisfies

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Equivalently,

$$g(n) \geq \left\lceil \frac{7n}{27} \right\rceil.$$

The proof uses Swanepoel's structural theorem and Lemmas 8–10, specialized to $m = 7$, and supplies the finite Euclidean incidence ledger needed to remove the unique possible $m = 7$ broken-lattice obstruction.

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1. Definitions and theorem

Definition 1 (Euclidean minimum-distance graph). *A graph G is a Euclidean minimum-distance graph if its vertices are represented by a finite set $P \subset \mathbb{R}^2$ such that*

$$|x - y| \geq 1 \quad (x, y \in P, x \neq y),$$

and two vertices are adjacent exactly when their Euclidean distance is 1.

Definition 2 (Open neighbourhood). *For $S \subseteq V(G)$, let*

$$N(S) = \{v \in V(G) \setminus S : v \text{ has a neighbour in } S\}$$

be the open neighbourhood of S .

Definition 3. *Let*

$$g(n) = \min \alpha(G),$$

where the minimum is over all Euclidean minimum-distance graphs on n vertices.

Theorem 1. *Every Euclidean minimum-distance graph G satisfies*

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Consequently,

$$\alpha(G) \geq \left\lceil \frac{7|V(G)|}{27} \right\rceil$$

and

$$g(n) \geq \left\lceil \frac{7n}{27} \right\rceil.$$

2. Published Swanepoel input

The proof imports exactly the following Swanepoel facts from [4]. The notation follows Swanepoel's broken-lattice proof.

Proposition 1 (Swanepoel package). *Let $m \geq 5$, and let G be a smallest counterexample to*

$$\alpha(G) \geq \frac{m}{4m-1}|V(G)|.$$

Then the following hold.

<1>1. *There is a nonconcave boundary arc*

$$p_0, p_1, \dots, p_{2m-3}$$

with total turn

$$\tau_1 + \dots + \tau_{2m-3} < \frac{\pi}{3}.$$

Moreover,

$$\deg(p_0) = 3,$$

and

$$\deg(p_i) = 4 \quad (1 \leq i \leq 2m-3).$$

For each edge $p_i p_{i+1}$ there is a unique common neighbour q_i .

<1>2. For $1 \leq i \leq 2m - 5$, Swanepoel's Lemma 8 gives exactly two neighbours r_i, s_i of q_i outside

$$\{p_i, p_{i+1}, q_{i-1}, q_{i+1}\},$$

with unavailable endpoint q -terms omitted.

<1>3. If

$$s_i = r_{i+1} \quad (i = a, a + 1, a + 2),$$

then

$$a \geq 2m - 10.$$

This is the triple obstruction from Swanepoel's Theorem 4 / Lemma 9.

<1>4. The proof of Swanepoel's Lemma 9 also gives the cap

$$|N(s_{a+1}) \setminus \{s_a, s_{a+2}, q_{a+1}, q_{a+2}\}| \leq 2$$

under the same three consecutive equalities.

<1>5. In the Euclidean plane, Swanepoel's Lemma 10 gives

$$s_i = r_{i+1} \quad (1 \leq i \leq 2m - 6)$$

for all but at most one index. If $s_i \neq r_{i+1}$, then

$$\tau_i + 2\tau_{i+1} + \tau_{i+2} \geq \frac{\pi}{3}.$$

Remark 1. The cap in item 4 is used below only for t_5, t_6, t_7 . It is a fact extracted from the proof of Swanepoel's Lemma 9, not from a separate theorem statement. This distinction is important for the bookkeeping in the finite ledger. The other background references are Pach–Toth's upper construction [2], Csizmadia's earlier lower bound [1], and Pollack's one-page note [3].

3. Minimal-counterexample deletion criterion

Lemma 1 (Deletion criterion). Let G be a smallest counterexample to

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

If $S \subseteq V(G)$ is independent and

$$|N(S)| \leq \left\lfloor \frac{20|S|}{7} \right\rfloor,$$

then G is not a counterexample.

Proof <1>1. Let $n = |V(G)|$.

<1>2. Put

$$G' = G - (S \cup N(S)).$$

<1>3. By minimality,

$$\alpha(G') \geq \frac{7}{27}|V(G')|.$$

<1>4. Since S has no edge to G' , an independent set in G' can be joined with S . Hence

$$\alpha(G) \geq |S| + \alpha(G').$$

<1>5. Also,

$$|V(G')| = n - |S| - |N(S)|.$$

<1>6. Therefore

$$\begin{aligned} \alpha(G) &\geq |S| + \frac{7}{27}(n - |S| - |N(S)|) \\ &= \frac{7n}{27} + \frac{20|S| - 7|N(S)|}{27}. \end{aligned}$$

<1>7. If

$$|N(S)| \leq \frac{20|S|}{7},$$

then

$$20|S| - 7|N(S)| \geq 0.$$

<1>8. Hence

$$\alpha(G) \geq \frac{7n}{27},$$

contradiction. □

4. The $m=7$ broken-lattice setup

Assume, for contradiction, that G is a smallest counterexample to Theorem 1. Apply Proposition 1 with $m = 7$. Then

$$2m - 3 = 11,$$

$$2m - 6 = 8,$$

and

$$2m - 10 = 4.$$

Thus the boundary arc is

$$p_0, p_1, \dots, p_{11}.$$

Lemma 2 (Unique $m = 7$ break). *The unique possible failed equality among*

$$s_i = r_{i+1} \quad (1 \leq i \leq 8)$$

is

$$s_3 \neq r_4.$$

Consequently,

$$s_i = r_{i+1} \quad i = 1, 2, 4, 5, 6, 7, 8.$$

Proof <1>1. By Proposition 1, at most one equality among

$$s_i = r_{i+1} \quad (1 \leq i \leq 8)$$

fails.

<1>2. By the triple obstruction, a true triple beginning at a forces

$$a \geq 4.$$

<1>3. Hence true triples beginning at $a = 1, 2, 3$ are impossible.

<1>4. These triples are

$$(1, 2, 3), \quad (2, 3, 4), \quad (3, 4, 5).$$

<1>5. A single failed equality must hit all three triples.

<1>6. The only common index of these three triples is 3.

<1>7. Hence the unique possible failed equality is

$$s_3 \neq r_4.$$

□

Whenever $s_i = r_{i+1}$, write

$$t_i = s_i = r_{i+1}.$$

Thus t_i is defined for

$$i = 1, 2, 4, 5, 6, 7, 8.$$

5. Coordinate conventions and elementary geometry

Identify $\mathbb{R}^2$ with $\mathbb{C}$, and put

$$\omega = e^{i\pi/3}.$$

Let

$$e_i = p_{i+1} - p_i, \quad |e_i| = 1.$$

Orient the boundary arc so that

$$q_i = p_i + \omega e_i.$$

Use the turn convention

$$e_{i+1} = e_i e^{-i\tau_{i+1}}, \quad \tau_{i+1} \geq 0.$$

The total turn is

$$\tau_1 + \cdots + \tau_{11} < \frac{\pi}{3}.$$

Lemma 3 (Degree bound). *Every vertex of G has degree at most 6.*

Proof <1>1. Let $v \in V(G)$.

<1>2. Every neighbour of v lies on the unit circle centered at v .

<1>3. If $x, y \in N(v)$ are distinct, then $|x - y| \geq 1$.

<1>4. Let θ be the smaller angular separation between x and y about v . Then

$$|x - y| = 2 \sin(\theta/2).$$

<1>5. Hence

$$2 \sin(\theta/2) \geq 1,$$

so

$$\theta \geq \frac{\pi}{3}.$$

<1>6. At most six directions around a circle can have pairwise angular separation at least $\pi/3$.

<1>7. Therefore $\deg(v) \leq 6$. □

Lemma 4 (Equality-strip coordinates). *If $s_i = r_{i+1}$, then*

$$t_i = p_i + \omega(e_i + e_{i+1}).$$

Moreover,

$$q_i q_{i+1} \in E(G) \iff \tau_{i+1} = 0,$$

and

$$t_i t_{i+1} \in E(G) \iff \tau_{i+1} + \tau_{i+2} = 0.$$

Proof <1>1. Since $q_i = p_i + \omega e_i$, the two intersections of the unit circles centered at q_i and q_{i+1} are p_{i+1} and

$$q_i + q_{i+1} - p_{i+1}.$$

<1>2. Therefore, on an equality strip,

$$\begin{aligned} t_i &= q_i + q_{i+1} - p_{i+1} \\ &= (p_i + \omega e_i) + (p_{i+1} + \omega e_{i+1}) - p_{i+1} \\ &= p_i + \omega(e_i + e_{i+1}). \end{aligned}$$

<1>3. Next,

$$\begin{aligned} q_{i+1} - q_i &= e_i + \omega e_{i+1} - \omega e_i \\ &= (1 - \omega)e_i + \omega e_{i+1} \\ &= \bar{\omega}e_i + \omega e_{i+1}. \end{aligned}$$

<1>4. The two summands are unit vectors. Their angle is

$$\frac{2\pi}{3} - \tau_{i+1}.$$

<1>5. Since $0 \leq \tau_{i+1} < \pi/3$, the sum has length 1 exactly when this angle is $2\pi/3$, i.e. exactly when $\tau_{i+1} = 0$.

<1>6. Thus

$$q_i q_{i+1} \in E(G) \iff \tau_{i+1} = 0.$$

<1>7. Similarly,

$$\begin{aligned} t_{i+1} - t_i &= p_{i+1} + \omega(e_{i+1} + e_{i+2}) - p_i - \omega(e_i + e_{i+1}) \\ &= e_i - \omega e_i + \omega e_{i+2} \\ &= \bar{\omega}e_i + \omega e_{i+2}. \end{aligned}$$

<1>8. The angle between these two unit summands is

$$\frac{2\pi}{3} - (\tau_{i+1} + \tau_{i+2}).$$

<1>9. Hence

$$t_i t_{i+1} \in E(G) \iff \tau_{i+1} + \tau_{i+2} = 0.$$

□

Corollary 1. *If $t_4 t_5 \in E(G)$, then*

$$\tau_5 = \tau_6 = 0.$$

After this, the following cases are exhaustive:

$$\tau_7 > 0,$$

$$\tau_7 = 0, \quad \tau_8 + \tau_9 > 0,$$

and

$$\tau_7 = \tau_8 = \tau_9 = 0.$$

Proof. By Lemma 4,

$$t_4 t_5 \in E(G) \iff \tau_5 + \tau_6 = 0.$$

Since $\tau_5, \tau_6 \geq 0$, this is equivalent to $\tau_5 = \tau_6 = 0$. The displayed trichotomy is the complete trichotomy for the nonnegative later turns τ_7, τ_8, τ_9 . □

Lemma 5 (Forward cone). *Rotate so that $e_a = 1$. For every $j \geq a$, there exists $\phi_j \in [0, \pi/3)$ such that*

$$e_j = e^{-i\phi_j}.$$

Consequently,

$$\Re(e_j) > \frac{1}{2},$$

and

$$\Re(\omega e_j) \geq \frac{1}{2}.$$

Proof. The total turn is $< \pi/3$, and all turns are nonnegative. Therefore every later edge direction is obtained from e_a by a clockwise turn $\phi_j \in [0, \pi/3)$. Thus $e_j = e^{-i\phi_j}$. Hence $\Re(e_j) = \cos \phi_j > 1/2$. Also $\omega e_j = e^{i(\pi/3 - \phi_j)}$, with $0 \leq \pi/3 - \phi_j \leq \pi/3$, so $\Re(\omega e_j) \geq 1/2$. □

Lemma 6 (Triangular-lattice units). *Let*

$$z = a + b\omega, \quad a, b \in \mathbb{Z}.$$

Then

$$|z|^2 = a^2 + ab + b^2.$$

The equation $|z| = 1$ has exactly the six solutions

$$z \in \{\pm 1, \pm \omega, \pm \omega^2\}.$$

Proof. Since $\omega + \bar{\omega} = 1$,

$$\begin{aligned} |a + b\omega|^2 &= (a + b\omega)(a + b\bar{\omega}) \\ &= a^2 + ab(\omega + \bar{\omega}) + b^2 \\ &= a^2 + ab + b^2. \end{aligned}$$

If $a^2 + ab + b^2 = 1$, then

$$a^2 + ab + b^2 = \left(a + \frac{b}{2}\right)^2 + \frac{3}{4}b^2,$$

so $|b| \leq 1$. If $b = 0$, then $a = \pm 1$. If $b = 1$, then $a^2 + a = 0$, so $a = 0$ or $a = -1$, giving ω or ω^2 . If $b = -1$, then $a^2 - a = 0$, so $a = 0$ or $a = 1$, giving $-\omega$ or $-\omega^2$. These are exactly the six listed vectors. $\square$

6. Boundary and endpoint neighbourhoods

Lemma 7. For $1 \leq i \leq 10$,

$$N(p_i) = \{p_{i-1}, p_{i+1}, q_{i-1}, q_i\}.$$

Also, for some vertex p_{-1} ,

$$N(p_0) = \{p_{-1}, p_1, q_0\},$$

and

$$N(p_{11}) \subseteq \{p_{10}, q_{10}\} \cup U_{11}, \quad |U_{11}| \leq 2.$$

Proof. For $1 \leq i \leq 10$, Proposition 1 gives $\deg(p_i) = 4$. The four displayed vertices are neighbours of p_i . We prove that they are distinct.

First,

$$p_{i+1} - p_{i-1} = e_{i-1} + e_i.$$

If $p_{i+1} = p_{i-1}$, then $e_i = -e_{i-1}$, requiring a turn of π , impossible because total turn is $< \pi/3$.

Next,

$$q_{i-1} = p_{i-1} + \omega e_{i-1} = p_i + (\omega - 1)e_{i-1} = p_i + \omega^2 e_{i-1},$$

so $q_{i-1} \neq p_i$. Similarly $q_i = p_i + \omega e_i \neq p_i$. Also $q_{i-1} - p_{i-1} = \omega e_{i-1} \neq 0$ and $q_i - p_{i+1} = \omega^2 e_i \neq 0$.

If $q_i = p_{i-1}$, then $\omega e_i = -e_{i-1}$, so $e_i = e^{2\pi i/3} e_{i-1}$, a counterclockwise turn $2\pi/3$, impossible. If $q_{i-1} = p_{i+1}$, then $e_i = \omega^2 e_{i-1}$, again a counterclockwise turn $2\pi/3$, impossible. If $q_{i-1} = q_i$, then $e_i = \omega e_{i-1}$, a counterclockwise turn $\pi/3$, impossible.

Thus the four displayed neighbours are distinct. Since $\deg(p_i) = 4$, they are exactly all neighbours of p_i .

For p_0 , Proposition 1 gives $\deg(p_0) = 3$. The vertices p_1, q_0 are neighbours, so let p_{-1} be the third neighbour. Then $N(p_0) = \{p_{-1}, p_1, q_0\}$.

For p_{11} , Proposition 1 gives $\deg(p_{11}) = 4$. The vertices p_{10}, q_{10} are neighbours, so at most two further neighbours exist. Hence the stated inclusion. $\square$

Lemma 8 (Endpoint q -caps).

$$N(q_1) \subseteq \{p_1, p_2, q_0, q_2, t_1\} \cup U_1, \quad |U_1| \leq 1.$$

Also,

$$N(q_9) \subseteq \{p_9, p_{10}, q_8, q_{10}, t_8\} \cup U_9, \quad |U_9| \leq 1.$$

Proof. By Swanepoel Lemma 8, q_1 has exactly two extra neighbours outside $\{p_1, p_2, q_0, q_2\}$. One is $t_1 = s_1 = r_2$. Thus at most one extra neighbour remains, proving the first inclusion.

Similarly, q_9 has exactly two extra neighbours outside $\{p_9, p_{10}, q_8, q_{10}\}$. One is $t_8 = s_8 = r_9$. Thus at most one extra neighbour remains, proving the second inclusion. $\square$

7. Separation lemmas

Lemma 9 (Boundary-boundary separation). *If $b - a \geq 2$, then*

$$|p_b - p_a| > 1.$$

Proof. If $b = a + 2$, then $p_b - p_a = e_a + e_{a+1}$, so

$$|p_b - p_a| = 2 \cos(\tau_{a+1}/2) > \sqrt{3} > 1.$$

If $b - a \geq 3$, then $p_b - p_a = e_a + e_{a+1} + \dots + e_{b-1}$. Project onto e_a . The first three terms contribute 1, $> 1/2$, and $> 1/2$, so the projection is $> 2 > 1$. Hence $|p_b - p_a| > 1$. $\square$

Lemma 10 (Boundary-to- q separation). *If $a \notin \{b, b + 1\}$, then*

$$|p_a - q_b| > 1.$$

Proof. If $a < b$, then $q_b - p_a = e_a + \dots + e_{b-1} + \omega e_b$. Project onto e_a . The e_a term contributes 1, and ωe_b contributes at least $1/2$. Hence the projection is > 1 .

If $a > b + 1$, then

$$p_a - q_b = e_b + \dots + e_{a-1} - \omega e_b = \bar{\omega} e_b + e_{b+1} + \dots + e_{a-1}.$$

For $a = b + 2$, the angle between $\bar{\omega} e_b$ and e_{b+1} is $\pi/3 - \tau_{b+1} \in [0, \pi/3]$, giving squared length at least 3. For $a \geq b + 3$, project onto e_b : the first three terms $\bar{\omega} e_b, e_{b+1}, e_{b+2}$ contribute $1/2, > 1/2, > 1/2$, so the projection is $> 3/2 > 1$. $\square$

Lemma 11 (Boundary-to- t separation). *Let*

$$t_b = p_b + \omega(e_b + e_{b+1}).$$

If p_a is not one of the explicitly displayed local neighbours of t_b , then

$$|p_a - t_b| > 1.$$

Proof. If $a < b$, then

$$t_b - p_a = e_a + \dots + e_{b-1} + \omega e_b + \omega e_{b+1}.$$

Projecting onto e_a , the first e -term contributes 1 and the two ωe terms each contribute at least $1/2$, so the projection is at least $2 > 1$.

If $a = b$, then $t_b - p_b = \omega(e_b + e_{b+1})$, whose length is $2 \cos(\tau_{b+1}/2) > \sqrt{3} > 1$.

If $a = b + 1$, then

$$p_{b+1} - t_b = \bar{\omega} e_b - \omega e_{b+1}.$$

Writing $e_{b+1} = e_b e^{-i\tau_{b+1}}$, this squared length is

$$2 - 2 \cos\left(\frac{2\pi}{3} - \tau_{b+1}\right) > 1,$$

because $0 \leq \tau_{b+1} < \pi/3$.

If $a = b + 2$, then

$$p_{b+2} - t_b = \bar{\omega}(e_b + e_{b+1}),$$

which has length $> \sqrt{3}$.

If $a > b + 2$, then

$$p_a - t_b = \bar{\omega}e_b + \bar{\omega}e_{b+1} + e_{b+2} + \cdots + e_{a-1}.$$

Project onto e_{b+2} . The terms e_{b+2} , $\bar{\omega}e_{b+1}$, and $\bar{\omega}e_b$ contribute at least 1, $1/2$, and $1/2$, respectively. Later terms have nonnegative projection. Hence the projection is at least $2 > 1$. $\square$

Lemma 12 ($q_7 t_5$ separation). *If $t_4 t_5 \notin E(G)$, then*

$$|q_7 - t_5| > 1.$$

Proof. We prove $|q_7 - t_5| \geq \sqrt{3}$. Since

$$q_7 = p_7 + \omega e_7$$

and

$$t_5 = p_5 + \omega(e_5 + e_6),$$

we have

$$q_7 - t_5 = e_5 + e_6 + \omega e_7 - \omega e_5 - \omega e_6 = \bar{\omega}(e_5 + e_6) + \omega e_7.$$

Rotate so $e_5 = 1$, and write

$$e_6 = e^{-ia}, \quad e_7 = e^{-i(a+b)},$$

with $a, b \geq 0$ and $a + b < \pi/3$. Multiplying by ω preserves length, so

$$|q_7 - t_5| = |1 + e^{-ia} + \omega^2 e^{-i(a+b)}|.$$

Let

$$F(a, b) = 1 + e^{-ia} + \omega^2 e^{-i(a+b)}.$$

Then

$$|F(a, b)|^2 = 3 + 2H(a, b),$$

where

$$H(a, b) = \cos a + \cos\left(\frac{2\pi}{3} - b\right) + \cos\left(\frac{2\pi}{3} - a - b\right).$$

For fixed a ,

$$\frac{\partial H}{\partial b} = \sin\left(\frac{2\pi}{3} - b\right) + \sin\left(\frac{2\pi}{3} - a - b\right) \geq 0,$$

because both arguments lie in $(\pi/3, 2\pi/3]$. Thus $H(a, b)$ is minimized at $b = 0$.

Now

$$\begin{aligned} H(a, 0) &= \cos a - \frac{1}{2} + \cos\left(\frac{2\pi}{3} - a\right) \\ &= \frac{1}{2} \cos a - \frac{1}{2} + \frac{\sqrt{3}}{2} \sin a. \end{aligned}$$

Therefore

$$2H(a, 0) = \cos a + \sqrt{3} \sin a - 1 = 2 \cos(a - \pi/3) - 1 \geq 0,$$

because $0 \leq a < \pi/3$. Hence $H(a, b) \geq 0$, so $|F(a, b)|^2 \geq 3$. $\square$

8. Cap lemmas

Lemma 13 (Swanepoel t -caps).

$$|N(t_5) \setminus \{t_4, t_6, q_5, q_6\}| \leq 2,$$

$$|N(t_6) \setminus \{t_5, t_7, q_6, q_7\}| \leq 2,$$

and

$$|N(t_7) \setminus \{t_6, t_8, q_7, q_8\}| \leq 2.$$

Proof. Apply the Swanepoel Lemma 9 cap from Proposition 1 to the equality triples $(4, 5, 6)$, $(5, 6, 7)$, and $(6, 7, 8)$.

For $t_5 = s_5$, the cap set is $\{s_4, s_6, q_5, q_6\} = \{t_4, t_6, q_5, q_6\}$. For $t_6 = s_6$, the cap set is $\{s_5, s_7, q_6, q_7\} = \{t_5, t_7, q_6, q_7\}$. For $t_7 = s_7$, the cap set is $\{s_6, s_8, q_7, q_8\} = \{t_6, t_8, q_7, q_8\}$. $\square$

Lemma 14 (Fully flat port cap). *In the fully flat branch, every extra neighbour of t_i , $i = 5, 6, 7$, outside the strip-neighbour set lies in one closed 60° -port. If the port is not saturated, it contributes at most one extra neighbour. If one endpoint is already present and the port is not saturated, it contributes no further extra neighbour. Consecutive saturated ports share one endpoint.*

Proof. Place $t_i = (0, 0)$. The four forced strip-neighbour directions are

$$0^\circ, \quad 180^\circ, \quad 240^\circ, \quad 300^\circ.$$

An extra unit neighbour must be separated by at least 60° from all four directions. The forbidden arcs are

$$(-60^\circ, 60^\circ),$$

$$(120^\circ, 240^\circ),$$

$$(180^\circ, 300^\circ),$$

and

$$(240^\circ, 360^\circ).$$

The only remaining arc is $[60^\circ, 120^\circ]$. Thus every extra neighbour lies in that port.

If two extra neighbours lie in that port, their angular separation is at most 60° . Since their mutual distance must be at least 1, their angular separation is exactly 60° , so they are the two endpoints.

If one endpoint is already present and another extra neighbour existed, then either it would be the other endpoint, saturating the port, or it would be an interior point at angular separation $< 60^\circ$ from the endpoint, hence at distance < 1 , impossible. Therefore no further extra neighbour exists in that case.

Finally,

$$t_i + \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = t_{i+1} + \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

Thus consecutive saturated ports share one endpoint. $\square$

Lemma 15 (Branch-qualified degree caps). *In the fully flat branch, if the t_7 -port is saturated, then*

$$N(t_8) \subseteq \{q_8, q_9, t_7, x_7\} \cup U_8, \quad |U_8| \leq 2.$$

In the all-saturated branch,

$$N(x_6) \subseteq \{t_6, t_7, x_5, x_7\} \cup V_6, \quad |V_6| \leq 2.$$

In the branch where t_6, t_7 are saturated but t_5 is not,

$$N(x_6) \subseteq \{t_6, t_7, y_6, x_7\} \cup V_6, \quad |V_6| \leq 2.$$

Proof. In fully flat coordinates,

$$t_8 = 3 + 2\omega.$$

If the t_7 -port is saturated, then $x_7 = 2 + 3\omega$. The vertices

$$q_8 = 3 + \omega, \quad q_9 = 4 + \omega, \quad t_7 = 2 + 2\omega, \quad x_7 = 2 + 3\omega$$

are at distance 1 from t_8 , since their differences from t_8 are $-\omega$, $1 - \omega$, -1 , and $-1 + \omega$. Since $\deg(t_8) \leq 6$, there are at most two further neighbours.

For $x_6 = 1 + 3\omega$, in the all-saturated branch the vertices

$$t_6 = 1 + 2\omega, \quad t_7 = 2 + 2\omega, \quad x_5 = 3\omega, \quad x_7 = 2 + 3\omega$$

are at distance 1 from x_6 . Their differences are $-\omega$, $1 - \omega$, -1 , and 1 . Hence the degree-six bound gives at most two further neighbours.

If t_6, t_7 are saturated but t_5 is not, then the actual left endpoint adjacent to x_6 is $y_6 = 3\omega$, not an actual saturated-port vertex x_5 . The four known neighbours of x_6 are t_6, t_7, y_6, x_7 , again with differences $-\omega$, $1 - \omega$, -1 , and 1 . Hence the degree-six bound gives at most two further neighbours. $\square$

Lemma 16 (Neighbourhood inclusion bookkeeping). *Let S be independent. Suppose that for every $v \in S$,*

$$N(v) \subseteq A \cup C_1 \cup \dots \cup C_r \cup S.$$

Then

$$N(S) \subseteq A \cup C_1 \cup \dots \cup C_r.$$

Consequently,

$$|N(S)| \leq |A| + \sum_{j=1}^r |C_j|.$$

Proof. If $x \in N(S)$, then $x \notin S$ and $x \in N(v)$ for some $v \in S$. By the assumed inclusion, $x \in A \cup C_1 \cup \dots \cup C_r \cup S$. Since $x \notin S$, $x \in A \cup C_1 \cup \dots \cup C_r$. $\square$

9. The t_4t_5 -absent branch

Lemma 17. *If $t_4t_5 \notin E(G)$, then*

$$S_0 = \{p_0, p_2, p_4, p_6, p_9, q_7, t_5\}$$

is independent and

$$|N(S_0)| \leq 20.$$

Proof. The selected boundary vertices are mutually nonadjacent by Lemma 9. They are nonadjacent to q_7 by Lemma 10, and nonadjacent to t_5 by Lemma 11. Finally, $q_7 t_5 \notin E(G)$ by Lemma 12. Hence S_0 is independent.

The selected boundary vertices contribute only

$$p_{-1}, p_1, p_3, p_5, p_7, p_8, p_{10}, q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_8, q_9.$$

The vertex q_7 contributes only

$$p_7, p_8, q_6, q_8, t_6, t_7.$$

The vertex t_5 has displayed possible neighbours t_4, t_6, q_5, q_6 , but $t_4 t_5$ is absent. Thus it contributes t_6, q_5, q_6 plus at most two further neighbours by Lemma 13.

Let

$$A_0 = \{p_{-1}, p_1, p_3, p_5, p_7, p_8, p_{10}, q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_8, q_9, t_6, t_7\}.$$

Then $|A_0| \leq 18$, and

$$N(S_0) \subseteq A_0 \cup C_5, \quad |C_5| \leq 2.$$

Therefore

$$|N(S_0)| \leq 18 + 2 = 20.$$

□

10. The t4t5-present branch with tau7 positive

Lemma 18. *Assume $t_4 t_5 \in E(G)$ and $\tau_7 > 0$. Then*

$$S_1 = \{p_6, p_8, q_4, t_5, t_6, t_7\}$$

is independent and

$$|N(S_1)| \leq 17.$$

Proof. By Corollary 1, $\tau_5 = \tau_6 = 0$. The boundary pair p_6, p_8 is nonadjacent by Lemma 9. They are nonadjacent to q_4 by Lemma 10, and nonadjacent to t_5, t_6, t_7 by Lemma 11.

The unique break inequality gives

$$\tau_3 + 2\tau_4 + \tau_5 \geq \frac{\pi}{3}.$$

Since $\tau_5 = 0$ and the total turn is $< \pi/3$, $\tau_4 > 0$. Hence $q_3 q_4 \notin E(G)$. By Swanepoel Lemma 8,

$$N(q_4) \subseteq \{p_4, p_5, q_3, q_5, r_4, s_4\}.$$

Since $q_3 q_4$ is absent and $s_4 = t_4$,

$$N(q_4) \subseteq \{p_4, p_5, q_5, r_4, t_4\}.$$

Normalize $p_4 = 0$ and $e_4 = e_5 = e_6 = 1$. Let $a = \tau_7 > 0$, $b = \tau_8 \geq 0$, with $a + b < \pi/3$. Then

$$q_4 = \omega,$$

$$t_5 = 1 + 2\omega,$$

$$t_6 = 2 + \omega(1 + e^{-ia}),$$

$$t_7 = 3 + \omega(e^{-ia} + e^{-i(a+b)}).$$

Now $|t_5 - q_4| = |1 + \omega| = \sqrt{3} > 1$. Also $t_6 - q_4 = 2 + \omega e^{-ia}$ has real part $2 + \cos(\pi/3 - a) > 1$, so $|t_6 - q_4| > 1$. Likewise $t_7 - q_4 = 3 - \omega + \omega e^{-ia} + \omega e^{-i(a+b)}$ has real part $3 - 1/2 + \cos(\pi/3 - a) + \cos(\pi/3 - a - b) > 1$.

By Lemma 4, $t_5 t_6$ is absent because $\tau_6 + \tau_7 > 0$, and $t_6 t_7$ is absent because $\tau_7 + \tau_8 > 0$. Finally,

$$t_7 - t_5 = 2\bar{\omega} + \omega(e^{-ia} + e^{-i(a+b)}).$$

Projection in the e_5 direction gives

$$2\cos(\pi/3) + \cos(\pi/3 - a) + \cos(\pi/3 - a - b) > 2.$$

Hence $|t_7 - t_5| > 1$. So S_1 is independent.

Let

$$A_1 = \{p_4, p_5, p_7, p_9, q_5, q_6, q_7, q_8, r_4, t_4, t_8\}.$$

Then $|A_1| \leq 11$. The non-capped neighbours lie in A_1 , and t_5, t_6, t_7 have Swanepoel caps C_5, C_6, C_7 , each of size at most 2. Thus

$$N(S_1) \subseteq A_1 \cup C_5 \cup C_6 \cup C_7,$$

and

$$|N(S_1)| \leq 11 + 2 + 2 + 2 = 17.$$

□

11. The t4t5-present branch with tau7 zero and a later turn

Lemma 19. *Assume $t_4 t_5 \in E(G)$, $\tau_7 = 0$, and $\tau_8 + \tau_9 > 0$. Then*

$$S_2 = \{p_0, p_3, p_5, p_8, q_1, q_6, t_7\}$$

is independent and

$$|N(S_2)| \leq 20.$$

Proof. By Corollary 1, $\tau_5 = \tau_6 = 0$. The selected boundary vertices are mutually nonadjacent by Lemma 9. They are nonadjacent to q_1, q_6 by Lemma 10, and nonadjacent to t_7 by Lemma 11.

Rotate so $e_1 = 1$. We have

$$q_6 - q_1 = e_1 + e_2 + e_3 + e_4 + e_5 + \omega e_6 - \omega e_1.$$

Therefore

$$\Re(q_6 - q_1) \geq 5 \cdot \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{5}{2} > 1.$$

Thus $q_1 q_6$ is absent.

Next,

$$t_7 - q_1 = e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + \omega e_7 + \omega e_8 - \omega e_1.$$

Therefore

$$\Re(t_7 - q_1) \geq 6 \cdot \frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{7}{2} > 1.$$

Thus $q_1 t_7$ is absent.

Since $\tau_6 = \tau_7 = 0$, $e_6 = e_7$, and

$$t_7 - q_6 = e_6 + \omega e_8.$$

The angle between e_6 and ωe_8 is $\pi/3 - \tau_8$, so

$$|t_7 - q_6|^2 = 2 + 2 \cos(\pi/3 - \tau_8) \geq 3 > 1.$$

Thus $q_6 t_7$ is absent, and S_2 is independent.

By Lemma 8,

$$N(q_1) \subseteq \{p_1, p_2, q_0, q_2, t_1\} \cup U_1, \quad |U_1| \leq 1.$$

Also,

$$N(q_6) \subseteq \{p_6, p_7, q_5, q_7, t_5, t_6\}.$$

Because $\tau_8 + \tau_9 > 0$, Lemma 4 gives $t_7 t_8 \notin E(G)$. By Lemma 13,

$$N(t_7) \subseteq \{q_7, q_8, t_6\} \cup C_7, \quad |C_7| \leq 2.$$

Let

$$A_2 = \{p_{-1}, p_1, p_2, p_4, p_6, p_7, p_9, q_0, q_2, q_3, q_4, q_5, q_7, q_8, t_1, t_5, t_6\}.$$

Then $|A_2| \leq 17$ and

$$N(S_2) \subseteq A_2 \cup U_1 \cup C_7.$$

Therefore

$$|N(S_2)| \leq 17 + 1 + 2 = 20.$$

□

12. Fully flat branch

Assume

$$\tau_7 = \tau_8 = \tau_9 = 0.$$

Normalize

$$p_5 = 0, \quad e_5 = e_6 = e_7 = e_8 = e_9 = 1.$$

Then

$$\begin{aligned} p_6 &= 1, & p_7 &= 2, & p_8 &= 3, & p_9 &= 4, & p_{10} &= 5, \\ q_5 &= \omega, & q_6 &= 1 + \omega, & q_7 &= 2 + \omega, & q_8 &= 3 + \omega, & q_9 &= 4 + \omega, \end{aligned}$$

and

$$t_5 = 2\omega, \quad t_6 = 1 + 2\omega, \quad t_7 = 2 + 2\omega, \quad t_8 = 3 + 2\omega.$$

Also,

$$p_{11} = 5 + e^{-i\theta}, \quad 0 \leq \theta < \frac{\pi}{3}.$$

When a port is saturated, write

$$x_i = t_i + \omega, \quad y_i = t_i + \omega^2.$$

Thus

$$x_5 = 3\omega, \quad x_6 = 1 + 3\omega, \quad x_7 = 2 + 3\omega,$$

and

$$y_6 = x_5, \quad y_7 = x_6.$$

Lemma 20. *If t_5 and t_7 are saturated, then t_6 is saturated.*

Proof. Saturation at t_5 supplies $x_5 = y_6$, one endpoint of the t_6 port. Saturation at t_7 supplies $x_6 = y_7$, the other endpoint. Hence both endpoints of the t_6 port are present, so t_6 is saturated. $\square$

Thus the possible non-all-saturated patterns are exactly

$$\emptyset, \quad t_5, \quad t_6, \quad t_7, \quad t_5t_6, \quad t_6t_7.$$

12.1. Six non-all-saturated rows

Row 1: no saturated port. Let

$$S_\emptyset = \{p_6, p_8, p_{11}, q_9, t_5, t_7\}.$$

The selected-pair distances are greater than 1: the lattice differences are not in the six unit vectors of Lemma 6, except where their length is displayed directly as 2 or $\sqrt{3}$; for pairs involving p_{11} ,

$$\Re(p_{11} - q_9) = 1 + \cos \theta - \frac{1}{2} > 1,$$

$$\Re(p_{11} - t_5) = 5 + \cos \theta - 1 = 4 + \cos \theta > 1,$$

$$\Re(p_{11} - t_7) = 3 + \cos \theta - 1 = 2 + \cos \theta > 1,$$

$$\Re(p_{11} - p_6) = 4 + \cos \theta > 1,$$

and

$$\Re(p_{11} - p_8) = 2 + \cos \theta > 1.$$

Thus $S_\emptyset$ is independent.

Let

$$A_\emptyset = \{p_5, p_7, p_9, p_{10}, q_5, q_6, q_7, q_8, q_{10}, t_4, t_6, t_8\}.$$

Then $|A_\emptyset| \leq 12$. With caps $|U_{11}| \leq 2$, $|U_9| \leq 1$, $|C_5| \leq 1$, and $|C_7| \leq 1$,

$$N(S_\emptyset) \subseteq A_\emptyset \cup U_{11} \cup U_9 \cup C_5 \cup C_7.$$

Hence

$$|N(S_\emptyset)| \leq 17.$$

Row 2: only t_5 saturated. Let

$$S_5 = \{p_5, p_8, p_{11}, q_6, q_9, t_7\}.$$

The selected-pair distances not involving p_{11} are again direct flat-coordinate checks. For p_{11} ,

$$\Re(p_{11} - p_5) = 5 + \cos \theta > 1,$$

$$\Re(p_{11} - p_8) = 2 + \cos \theta > 1,$$

$$\Re(p_{11} - q_6) = 4 + \cos \theta - \frac{1}{2} > 1,$$

$$\Re(p_{11} - q_9) = 1 + \cos \theta - \frac{1}{2} > 1,$$

and

$$\Re(p_{11} - t_7) = 3 + \cos \theta - 1 = 2 + \cos \theta > 1.$$

Thus S_5 is independent.

Let

$$A_5 = \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8\}.$$

Then $|A_5| \leq 13$, and

$$N(S_5) \subseteq A_5 \cup U_{11} \cup U_9 \cup C_7,$$

where $|U_{11}| \leq 2$, $|U_9| \leq 1$, and $|C_7| \leq 1$. Hence

$$|N(S_5)| \leq 17.$$

Row 3: only t_6 saturated. Use

$$S_6 = \{p_5, p_8, p_{11}, q_6, q_9, t_7\}.$$

It is independent by Row 2. Let

$$A_6 = \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8, x_6\}.$$

Then $|A_6| \leq 14$. The t_7 port is not saturated, and $x_6 = y_7$ is already supplied by the saturated t_6 port. By Lemma 14, there is no further t_7 port neighbour. Thus

$$N(S_6) \subseteq A_6 \cup U_{11} \cup U_9,$$

and

$$|N(S_6)| \leq 14 + 2 + 1 = 17.$$

Row 4: only t_7 saturated. Let

$$S_7 = \{p_0, p_3, p_6, q_1, q_4, t_5\}.$$

Boundary-boundary, boundary-to- q , and boundary-to- t nonadjacencies follow from Lemmas 9, 10, and 11. The remaining auxiliary pairs satisfy

$$\Re(q_4 - q_1) \geq \frac{3}{2} > 1,$$

$$\Re(t_5 - q_1) \geq \frac{5}{2} > 1,$$

and, after normalizing $p_4 = 0$, $q_4 = \omega$ and $t_5 = 1 + 2\omega$, so

$$|t_5 - q_4| = |1 + \omega| = \sqrt{3} > 1.$$

Thus S_7 is independent.

Let

$$A_7 = \{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_4, t_1, t_4, t_6\}.$$

Then $|A_7| \leq 15$, and

$$N(S_7) \subseteq A_7 \cup U_1 \cup C_5,$$

where $|U_1| \leq 1$ and $|C_5| \leq 1$. Hence

$$|N(S_7)| \leq 17.$$

Row 5: t_5, t_6 saturated. Use

$$S_{56} = \{p_5, p_8, p_{11}, q_6, q_9, t_7\}.$$

It is independent by Row 2. Let

$$A_{56} = \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8, x_6\}.$$

Then $|A_{56}| \leq 14$. The t_7 port is not saturated, and $x_6 = y_7$ is already supplied by the saturated t_6 port. Thus t_7 has no further port neighbour. Hence

$$N(S_{56}) \subseteq A_{56} \cup U_{11} \cup U_9,$$

and

$$|N(S_{56})| \leq 17.$$

Row 6: t_6, t_7 saturated. Let

$$S_{67} = \{p_6, p_9, q_7, t_5, t_8, x_6\}.$$

In flat coordinates,

$$p_6 = 1, \quad p_9 = 4, \quad q_7 = 2 + \omega, \quad t_5 = 2\omega, \quad t_8 = 3 + 2\omega, \quad x_6 = 1 + 3\omega.$$

The selected-pair differences are

$$\begin{aligned} & 3, \quad 1 + \omega, \quad 2\omega - 1, \quad 2 + 2\omega, \quad 3\omega, \quad -2 + \omega, \\ & 2\omega - 4, \quad -1 + 2\omega, \quad -3 + 3\omega, \quad \omega - 2, \quad 1 + \omega, \quad -1 + 2\omega, \\ & 3, \quad 1 + \omega, \quad -2 + \omega. \end{aligned}$$

None is one of $\pm 1, \pm\omega, \pm\omega^2$. Since all graph distances are at least 1, every selected-pair distance is > 1 . Hence S_{67} is independent.

Let

$$A_{67} = \{p_5, p_7, p_8, p_{10}, q_5, q_6, q_8, q_9, t_4, t_6, t_7, y_6, x_7\}.$$

Then $|A_{67}| \leq 13$. By Lemma 15,

$$N(t_8) \subseteq A_{67} \cup U_8, \quad |U_8| \leq 2,$$

and

$$N(x_6) \subseteq A_{67} \cup V_6, \quad |V_6| \leq 2.$$

Moreover, $y_6 = x_5$ is already an endpoint of the non-saturated t_5 port, so Lemma 14 gives no further t_5 port neighbour. Therefore

$$N(S_{67}) \subseteq A_{67} \cup U_8 \cup V_6,$$

and

$$|N(S_{67})| \leq 13 + 2 + 2 = 17.$$

13. All-saturated branch

Assume t_5, t_6, t_7 are all saturated. Define

$$S_3 = \{p_0, p_3, p_6, p_9, q_1, q_4, q_7, t_5, t_8, x_6\}.$$

Lemma 21. *The set S_3 is independent and*

$$|N(S_3)| \leq 28.$$

Proof. Boundary-boundary, boundary-to- q , and boundary-to- t nonadjacencies follow from Lemmas 9, 10, and 11.

For x_6 against selected boundary vertices, note that

$$x_6 = p_6 + 3\omega e_6.$$

Thus $|x_6 - p_6| = 3$. In flat coordinates, $x_6 - p_9 = -3 + 3\omega$, so $|x_6 - p_9| = 3$. For p_3 ,

$$x_6 - p_3 = e_3 + e_4 + e_5 + 3\omega e_6,$$

and, rotating so $e_3 = 1$,

$$\Re(x_6 - p_3) > 1 + \frac{1}{2} + \frac{1}{2} + 3 \cdot \frac{1}{2} = 3 > 1.$$

For p_0 ,

$$x_6 - p_0 = e_0 + e_1 + e_2 + e_3 + e_4 + e_5 + 3\omega e_6,$$

and, rotating so $e_0 = 1$,

$$\Re(x_6 - p_0) > 1 + 5 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = 5 > 1.$$

For pairs involving q_1 :

$$\Re(q_4 - q_1) \geq \frac{3}{2} > 1,$$

$$\Re(q_7 - q_1) \geq 3 > 1,$$

$$\Re(t_5 - q_1) \geq \frac{5}{2} > 1,$$

$$\Re(t_8 - q_1) \geq \frac{7}{2} > 1,$$

$$\Re(x_6 - q_1) \geq \frac{7}{2} > 1.$$

For q_4 , the break inequality gives $\tau_4 > 0$, so q_3q_4 is absent. In fully flat coordinates,

$$p_4 = -1, \quad q_4 = -1 + \omega,$$

and

$$q_7 = 2 + \omega, \quad t_5 = 2\omega, \quad t_8 = 3 + 2\omega, \quad x_6 = 1 + 3\omega.$$

Therefore

$$|q_7 - q_4| = 3 > 1,$$

$$|t_5 - q_4| = |1 + \omega| = \sqrt{3} > 1,$$

$$|t_8 - q_4| = |4 + \omega| > 1,$$

and

$$|x_6 - q_4| = |2 + 2\omega| = 2\sqrt{3} > 1.$$

Among q_7, t_5, t_8, x_6 , the differences are

$$\omega - 2, \quad 1 + \omega, \quad -1 + 2\omega, \quad 3, \quad 1 + \omega, \quad -2 + \omega.$$

None is a triangular-lattice unit vector, so all corresponding distances are > 1 . Hence S_3 is independent.

Now define

$$A_3 = \{p_{-1}, p_1, p_2, p_4, p_5, p_7, p_8, p_{10}, q_0, q_2, q_3, q_5, q_6, q_8, q_9, \\ r_4, t_1, t_4, t_6, t_7, x_5, x_7, y_5\}.$$

Then $|A_3| \leq 23$.

The neighbourhood inclusions are:

$$N(p_0), N(p_3), N(p_6), N(p_9) \subseteq A_3.$$

Also,

$$N(q_1) \subseteq A_3 \cup U_1, \quad |U_1| \leq 1.$$

Since q_3q_4 is absent and $s_4 = t_4$,

$$N(q_4) \subseteq A_3.$$

Also,

$$N(q_7) \subseteq A_3,$$

and, because t_5 is saturated,

$$N(t_5) \subseteq A_3.$$

By Lemma 15,

$$N(t_8) \subseteq A_3 \cup U_8, \quad |U_8| \leq 2,$$

and

$$N(x_6) \subseteq A_3 \cup V_6, \quad |V_6| \leq 2.$$

Thus

$$N(S_3) \subseteq A_3 \cup U_1 \cup U_8 \cup V_6,$$

and

$$|N(S_3)| \leq 23 + 1 + 2 + 2 = 28.$$

□

14. Final proof

Proof of Theorem 1. Let G be a smallest counterexample. By Lemma 1, every independent set S in G must violate

$$|N(S)| \leq \left\lfloor \frac{20|S|}{7} \right\rfloor.$$

By Lemma 2, the unique possible break is $s_3 \neq r_4$, and the equality vertices t_i exist for $i = 1, 2, 4, 5, 6, 7, 8$.

If $t_4t_5 \notin E(G)$, Lemma 17 gives an independent S_0 with

$$|S_0| = 7, \quad |N(S_0)| \leq 20 = \left\lfloor \frac{20 \cdot 7}{7} \right\rfloor,$$

contradiction.

Therefore $t_4t_5 \in E(G)$, and by Corollary 1,

$$\tau_5 = \tau_6 = 0.$$

If $\tau_7 > 0$, Lemma 18 gives an independent S_1 with

$$|S_1| = 6, \quad |N(S_1)| \leq 17 = \left\lfloor \frac{20 \cdot 6}{7} \right\rfloor,$$

contradiction.

If $\tau_7 = 0$ and $\tau_8 + \tau_9 > 0$, Lemma 19 gives an independent S_2 with

$$|S_2| = 7, \quad |N(S_2)| \leq 20 = \left\lfloor \frac{20 \cdot 7}{7} \right\rfloor,$$

contradiction.

Thus

$$\tau_7 = \tau_8 = \tau_9 = 0.$$

In the fully flat branch, if not all ports at t_5, t_6, t_7 are saturated, one of the six row sets above gives an independent set S with

$$|S| = 6, \quad |N(S)| \leq 17 = \left\lfloor \frac{20 \cdot 6}{7} \right\rfloor,$$

contradiction.

Hence all three ports are saturated. Lemma 21 gives an independent set S_3 with

$$|S_3| = 10, \quad |N(S_3)| \leq 28 = \left\lfloor \frac{20 \cdot 10}{7} \right\rfloor,$$

contradiction.

Every branch contradicts minimality. Therefore no smallest counterexample exists. Hence every Euclidean minimum-distance graph G satisfies

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Since $\alpha(G)$ is integer-valued,

$$\alpha(G) \geq \left\lceil \frac{7|V(G)|}{27} \right\rceil.$$

Taking the minimum over all n -vertex Euclidean minimum-distance graphs gives

$$g(n) \geq \left\lceil \frac{7n}{27} \right\rceil.$$

□

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